Sentiment Analysis of Comments Across Different Social Media Using IndoBERT

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***Abstract*—Social media platforms allow users to express emotions through comments. These comments contain sentiment, crucial for shaping public opinion, user behavior, and societal trends. This study investigates how sentiment patterns differ across platform and how IndoBERTweet can be utilized for sentiment classification. While previous studies have explored models like IndoBERT and IndoBERTweet for Indonesian language sentiment analysis, most focused on single platforms and lacked real-world cross-platform data evaluation. By conducting cross-platform sentiment analysis, this study addresses that gap. IndoBERTweet model was chosen to analyze and classify Indonesian-language user comments collected from Twitter, Instagram, and TikTok within the scope of the "#KaburAjaDulu" hashtag. Transfer learning was applied to a fine-tuned IndoBERTweet model for better sentiment classification, resulting in the Custom IndoBERTweet model. This new model outperformed the original, achieving higher accuracy. The sentiment analysis result highlighted significant differences in sentiment across platforms, influenced by platform-specific user demographics and content types. This finding offers practical applications for businesses and policymakers to effectively interpret and respond to online public sentiment.**

***Keywords*—social media, sentiment analysis, IndoBERTweet, transfer learning, informal Indonesian**

I. Introduction

In the digital age, social media platforms have become an integral part of daily communication, allowing people to share their opinions and moments on several topics, as well as for companies to advertise their products [1]–[3], instantaneously through posts with several formats such as text, audio, image, and video [4].

Each social media platform, such as Facebook, Instagram, Twitter, LinkedIn, and TikTok, has unique characteristics, attracting distinct user demographics due to their difference in the format focus [5]–[7].

Regardless of their differences, platforms depend on user behaviors. This is why user engagement is highly encouraged [8]. One of the most common forms of engagement is commenting on other user's posts. Properly processed, comments are invaluable to businesses, governments, and individuals [9].

One of the main approaches used in user comments research is sentiment analysis [10]. Sentiment analysis, also known as opinion analysis or opinion mining, is a solution that allows extracting summarized public opinion on any topic or context from large data sources [11].

Several studies in the past five years have explored sentiment analysis on social media platforms. By leveraging machine learning algorithms, sentiment analysis can classify text into categories such as positive, negative, or neutral, and provide valuable insights into public opinion, customer satisfaction, and market trends [12], [13].

However, there is still limited research on cross-platform sentiment analysis, as most studies focus on a single platform. Therefore, this research aims to analyze and compare sentiments in user-generated comments across three different platforms, which are Twitter, Instagram, and TikTok, each focusing on distinct content formats. The hashtag "#KaburAjaDulu" hashtag is chosen for data consistency.

The methodology involves data collection and cleaning, fine-tuning, transfer learning, sentiment classification, and results analysis. The fine-tuning process refers to finding the best model configuration, while transfer learning means building Custom IndoBERTweet by introducing new training datasets.

The paper is structured into five chapters: Introduction, Literature Review, Methodology, Results, and Conclusion. The Introduction discusses the significance of cross-platform sentiment analysis as well as the paper objectives, methodology, and outline. The Literature Review explores related theories and studies. The Methodology explains the detailed process of the model building and sentiment analysis, while the Results presents and discusses the findings. Finally, the Conclusion summarizes the whole research and suggests future improvements for similar studies.

II. Literature Review

This chapter reviews the importance of user-generated comments on social media, the way recent studies highlight the need to expand researches towards cross-platforms and informal languages, the technique chosen for this research, and the analysis of datasets used in prior studies as basis of collecting datasets.

This research explores sentiment analysis of Indonesian-language social media comments across Twitter, Instagram, and TikTok utilizing the deep learning model called IndoBERTweet.

A. Understanding Social Media Comments

Social media platforms have been growing rapidly, with people interacting through comments. Commenting is a way to express opinions, to discuss products, people, and issues, as well as to search for connection [10], [14]. These comments often convey various emotions reflected in the tone of the message, which can influence whether others perceive the associated topic positively or negatively [15], [16]. Furthermore, comments can assist businesses, governments, and institutions by revealing public perceptions—for example, helping a company better understand its consumers and develop more effective strategic marketing plans [17], [18].

Xu *et al.*, in their 2022 paper [19], emphasized the need to research various platforms on specific topics rather than focusing on single platforms. Omuya *et al.* in their 2023 paper [20] also recommended that future studies utilize larger and more varied user-generated content. Additionally, Alafwan *et al.*, in their 2023 paper, [10] recommended exploring less-studied platforms, such as Instagram and TikTok. Therefore, analyzing user comments on many platforms is critical in interpreting user engagement and public sentiment.

B. Expanding Across Platforms and Languages

Other than exploring less-studied platforms, Alafwan *et al.* also mentioned the importance of expanding studies to non-English texts, which was also advocated by Xu *et al.*. Furthermore, a 2024 paper by Indriani *et al.* [21] encouraged the exploration of shorter and informal texts commonly found on social media. However, as noted by Çelik *et al.* in their 2020 paper [22], the informal language and lack of adherence to grammar rules in social media comments can affect accuracy. A more recent study by Madyatmadja *et al.* which was published on 2024 [23] proposed that advanced preprocessing methodologies for particular language, such as Indonesian, could help improve accuracy. Altogether, these studies published over the past four years emphasize the need for research on a robust technique capable of handling varied informal Indonesian data across multiple platforms.

Platforms like Twitter, Instagram, and TikTok are widely used, including in Indonesia. Twitter, launched in 2006, is a microblogging platform with a 280-character limit, used mainly for news and public discourse. As of 2024, it has over 330 million users, primarily educated, higher-income men. Instagram, launched in 2010, focuses on photo and video sharing. It has attracted about 1.4 billion users, primarily unpartnered women around 18 to 24 years old. It features Stories, Reels, and an active influencer community, making it essential for brand marketing. TikTok, launched in 2016, is a short-video platform that rose to global popularity during the COVID-19 pandemic, especially among individuals under 24. In 2024, it has over 1 billion users globally. Its viral trends and short content make it highly engaging. [5], [7], [24]–[27] Given these differences in format and demographics, the three platforms are well-suited to analyze varied user comments on a relevant topic.

C. Utilizing Sentiment Analysis and Machine Learning

Sentiment analysis has emerged as a reliable method for analyzing user-generated content on social media. A 2022 paper by Xu *et al.* [19] highlighted that sentiment analysis is effective in extracting sentiment, emotion, and opinion from written language. Various approaches have been developed for sentiment analysis. In choosing a suitable strategy, past related studies were reviewed.

Early studies, such as those by Çelik *et al.* (2020) [22], employed classical models like Logistic Regression, Support Vector Machine, Random Forest, and Naive Bayes on datasets such as social media comments, Turkish tweets, and e-commerce reviews. These methods achieved moderate performance, typically ranging from 69% to 91% accuracy, depending on the dataset and feature engineering techniques (e.g., N-grams, TF-IDF).

Pinilla *et al.* (2021) [28] used NB, SVM, RF, and KNN on Facebook and call center texts, reporting accuracies up to 82%. Madyatmadja *et al.* (2024) [23] applied NB, SVM, and RF methods on Threads application reviews from the Play Store, achieving over 75% accuracy.

According to Xu *et al.* (2022) [19], deep learning methods like LSTM and transformer-based models like IndoBERT generally outperform traditional models like Logistic Regression and SVM, especially on informal and complex text. Notably, Indriani *et al.* (2024) [21] showed how IndoBERTweet—a specialized model of IndoBERT trained on Indonesian social media data—outperforms general-purpose multilingual models, such as IndoBERT and mBERT, proving it suitable for Indonesian informal text.

IndoBERT itself is one of the first pretrained monolingual BERT models for the Indonesian language [29], [30]. As its extension, IndoBERTweet was developed to process user-generated text more accurately.

Based on the review, IndoBERTweet would be more accurate in predicting Indonesian-language informal textual data compared to IndoBERT. This suggests that the IndoBERTwee, is suited for cross-platform sentiment analysis research focusing on Indonesian social media comments.

D. Analyzing Datasets Used in Related Works

Various studies on sentiment analysis draw from both real-world platforms and benchmark repositories. For example, Xu *et al.* [19] conducted a comprehensive review and found that

most datasets are either collected directly from platforms such as Twitter, Facebook, Reddit, and TripAdvisor, or sourced from well-established datasets like IMDB, Sentiment140, Amazon Product Reviews, and the Stanford Twitter Sentiment (STS) corpus. This focus reveals a gap—popular platforms like Instagram and TikTok remain underexplored.

In the Indonesian context, Hayati *et al.* [31] collected comments from TikTok accounts of two instant noodle brands, Indomie and Gaga. The data was collected through a crawling process using Python, resulting in 2861 data points after crawling. The preprocessing method used was case folding, which involves changing all letters to lowercase and removing punctuation marks, numbers, and empty characters. This data was used to determine public sentiment towards the two brands. The paper suggests that to increase the accuracy of sentiment grouping, future preprocessing could also include tokenizing, filtering, and stemming processes.

Meanwhile, Koto *et al.* [29] introduced the INDOLEM dataset, a comprehensive resource for Indonesian NLP tasks, including Twitter and hotel review data for sentiment analysis. The hotel review data was annotated at the aspect level, and preprocessing involved counting the proportion of positive and negative labels based on the majority sentiment, discarding reviews with ties. Other than that, the paper does not explicitly detail the preprocessing for each of the eight sub-datasets.

Most existing research concentrates on platforms like Twitter and Facebook. Although efforts on TikTok dataset like Hayati *et al.*'s are promising, more data collection across diverse platforms is needed. Expanding the platforms and Indonesian-language datasets would better support models like IndoBERTweet in cross-platform sentiment analysis.

III. Methodology

This chapter will cover the system architecture, followed by detailed steps and processes of collecting the user comments datasets, preprocessing the data, training the model, and classifying comments by sentiments.

A. System Architecture

The workflow of this research methodology is presented in the system architecture diagram (see Fig. 1). The gray nodes represent data collection, while the blue nodes show the process of how the data is used. Raw comments scraped from each platform are preprocessed before being manually labeled. A transfer learning approach is then employed by first fine-tuning the IndoBERTweet using the preprocessed data, creating a new model which better adapts to the cross-platform dataset. After that, sentiment classification is done using both the fine-tuned and the new model. Subsequently, the results of sentiment classification are analyzed to evaluate the performance of the custom model.

B. Data Collection

Data collection was done with acknowledgment and adherence to ethical practices concerning data scraping and handling. Comments were collected from publicly available posts

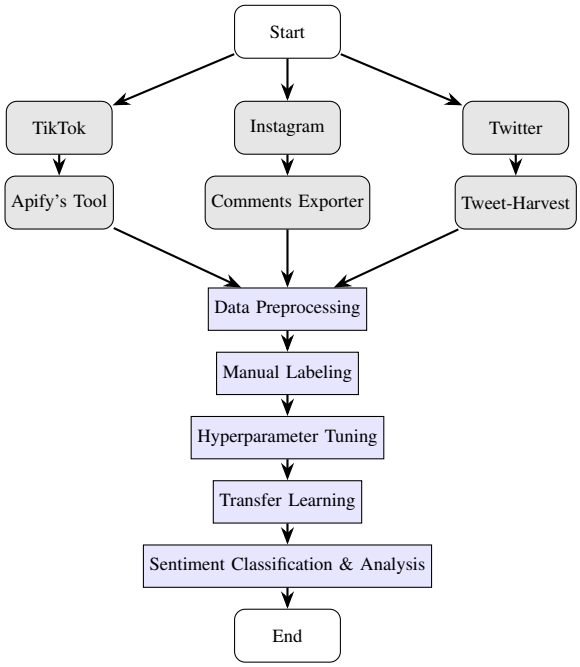

Fig. 1. System Architecture of Sentiment Analysis

without accessing private user data or accounts. Usernames and user mentions were removed during preprocessing to protect user privacy.

Our dataset initially consisted of 1,752 comments that were collected from Twitter, Instagram, and TikTok using web scraping. There were 665 comments from Twitter, 587 from Instagram, and 500 from TikTok. Twitter comments were scraped using Python, specifically the Tweet-Harvest [32] tool, while Instagram comments were extracted using the Chrome extension *Comments Exporter* [33]. For TikTok, the comments were scraped using TikTok data extractor on Apify [34].

To ensure consistency, the viral Indonesian hashtag "#KaburAjaDulu" (#JustRunAwayFirst) was used as a constant variable. Popularized in late January 2025 due to political events, it had been used on Twitter since September 2023 and later spread to platforms like Instagram and TikTok [35]–[37]. The hashtag reflects both positive and negative sentiments, supporting the cross platform analysis of the study [38].

C. Data Cleaning and Preprocessing

After collection, the raw datasets were cleaned to ensure that they are suitable for input to the IndoBERTweet model. The steps were lowercasing all the text and removing the unnecessary characters, such as numbers, whitespaces, punctuations, emojis, hashtags, mentions, and URLs. This data cleaning and preprocessing was written with Python 3.11.12 using libraries such as Pandas for data manipulation, NLTK for tokenization, and Regular Expressions for text cleaning. The data cleaning and preprocessing algorithm is provided in Algorithm 1.

An example of the process and its result is given in Table I, with the user ID censored for privacy. After cleaning and preprocessing, there are 640 rows of data left for Twitter,

Algorithm 1 Text Preprocessing Algorithm

Require: Dataset D with column `text`
Ensure: Cleaned text data in D

```
for each row i in D do
    Convert D[i].text to lowercase
    Replace all newline characters with a space
    Remove URLs starting with http or https
    Remove user mentions starting with @
    Remove all numeric digits
    Remove punctuation characters
    Remove leading and trailing whitespace
    Remove non-alphabetic characters and emojis
    Update D[i].text with the cleaned sentence
end for
for each row i in D do
    if D[i].text = ∅ then
        Delete row i from D
    end if
end for
```

527 rows for Instagram, and 492 rows for TikTok. In total, as many as 1,659 comments[1] are ready to be manually labeled and inputted into the pretrained IndoBERTweet sentiment classifier.

TABLE I
EXAMPLE OF DATA CLEANING AND PREPROCESSING

Source	Before	After
Instagram	@a***2 intinya kita hidup cuma 1x jadi jangan lakukan hal bodoh (*@a***2 the point is we only live 1 time so do not do stupid things*)	intinya kita hidup cuma x jadi jangan lakukan hal bodoh (*the point is we only live time so do not do stupid things*)
Twitter	@z*******97 malaysia sih paling make sense dari budaya dan bahasa yg ga begitu jauh dari indonesia (*@z*******97 malaysia makes the most sense from the culture and language which is not that far from Indonesia*)	malaysia sih paling make sense dari budaya dan bahasa yg ga begitu jauh dari indonesia (*malaysia makes the most sense from the culture and language which is not that far from Indonesia*)
TikTok	"nasionalisme.. ?? gaji gua 1,5 sebulan a.njjjjjj #kaburajadulu" (*"nationalism.. ?? my salary is 1.5 a month s.hitttttt #justrunawayfirst"*)	nasionalisme gaji gua sebulan anjjjjjj kaburajadulu (*nationalism my salary a month shitttttt justrunawayfirst*)

D. Manual Data Labeling Method and Process

The manual labeling process was carried out by carefully reviewing the same preprocessed comments based on their content one by one. Consistency is ensured by using the established criteria explained in Table II for the sentiment labeling. This manual labeling aims to better understand the characteristics of user sentiment across platforms, and will later be used to assess the performance of the IndoBERTweet models as well.

[1]https://github.com/stefani-gifta/cross-platform-dataset-2025

TABLE II
SENTIMENT LABELING CRITERIA

Sentiment	Characteristics	Examples
Negative	Dissatisfaction, frustration, criticism, hate speech, fear, complaints, sarcasm, and resentments.	• "saya cinta negara tapi negara ga cinta saya" (*i love my country, but my country doesn't love me*) • "kerja serius gaji bercanda #KaburAjaDulu" (*serious work, joking salary #JustRunAwayFirst*)
Positive	Pride, contentment, joy, excitement, praise, encouragement, and satisfaction.	• "indonesia terus maju dan bisa menjadi indonesia emas" (*indonesia continues to progress and can become a golden indonesia*) • "i m malaysian but i d always romanticising life in indonesia you guys are blessed with nature amp pretty sure within years you ll be a giant within sea"
Neutral	Agreement, questions, humorous statements without sarcasm, general facts with no strong emotions, and informative recommendation.	• "yaampun kak makasih banyak ya" (*oh my gosh sis, thank you so much, okay*) • "aamiin" (*aameen*)

E. Developing the Custom IndoBERTweet with Transfer Learning

To adapt the model to the cross-platforms comments, transfer learning was applied. Transfer learning involves training the fine-tuned IndoBERTweet model [39] on the newly preprocessed and manually labeled dataset. Transfer learning allows the model to capture the language patterns and slang in the comments of different platforms better.

The transfer learning method uses T4 GPU. The first step is loading the tokenizer and model of the fine-tuned IndoBERTweet [39]. After that, the three dataset comments from Twitter, Instagram, and TikTok that have been manually labeled is read using pandas library. The negative, neutral, and positive labeling are replaced by 0, 1, and 2. For the transfer learning, the dataset are concatted as one. This concatted dataset has the sentiment distribution of 943 negative, 488 neutral, and 225 positive.

Afterwards, the concatted dataset is split into three sets, which is training, validation, and test (60% train and 40% validation as well as test). Then, from the 40%, the dataset is split into 30% validation and 10% test. The next step is tokenizing the data to detect the comments label. After that, torch library is imported to prepare the data before transfer learning.

Hyperparameters were tuned via automated search using

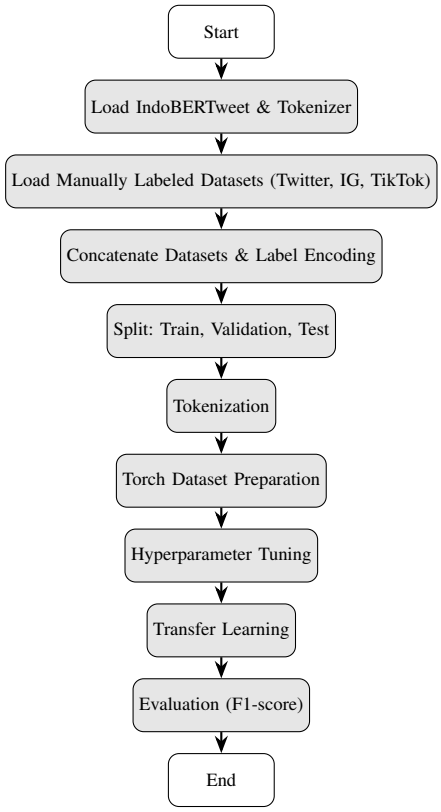

Fig. 2. Transfer Learning Process with IndoBERTweet for Cross-Platform Sentiment Analysis

the Optuna framework[2]. The hyperparameter explored were learning rate, number of fine-tuning epochs, weight decay, learning-rate decay strategy, and training batch size. All training runs were logged with Weights & Biases (W&B) for experiment tracking and reproducibility. Model performance was evaluated using F1-score. These metrics were computed with the sklearn.metrics library.

F. Data Tokenization and Sentiment Classification

The sentiment classification was done using two models on the same preprocessed dataset. The two models are the fine-tuned sentiment classifier [39] based on the IndoBERTweet base [40] and the custom model resulted after transfer learning of the fine-tuned IndoBERTweet. The use of IndoBERTweet enables an understanding of contextual Indonesian language [40] as it offers a more accurate interpretation of sentiment analysis on emotionally charged topics such as politics.

Prior to classification, the preprocessed data were tokenized. These tokenized inputs were then analyzed and classified as expressing positive, negative, or neutral sentiment. The sentiment classifications were done in Python 3.11.12, utilizing the HuggingFace Transformers 4.51.3 and PyTorch 2.6.0+cu124.

The code for sentiment classification using fine-tuned IndoBERTweet is given in Listing 1.

[2]https://optuna.org/

```python
from transformers import AutoTokenizer,
    AutoModelForSequenceClassification
import torch

# Load tokenizer and model
tokenizer = AutoTokenizer.from_pretrained("
    ridho2401/sa-tapera")
model = AutoModelForSequenceClassification.
    from_pretrained("ridho2401/sa-tapera")

# Function to predict sentiment
def predict_sentiment(text):
   inputs = tokenizer(text, return_tensors="pt
       ", truncation=True, padding=True,
       max_length=128)
   with torch.no_grad():
      outputs = model(**inputs)
   probs = torch.nn.functional.softmax(outputs
       .logits, dim=1)
   label = torch.argmax(probs, dim=1).item()

   label_map = {0: "positive", 1: "neutral",
       2: "negative"}
   return label_map[label]

data['sentiment'] = data['text'].apply(
    predict_sentiment)
```

Listing 1. Sentiment Classification using Python

The custom model sentiment prediction code is exactly the same, except for the label mapping. The original fine-tuned model has the mapping of 0 as positive, 1 as neutral, and 2 as negative, while the customized model switched the mapping of positive being 2 and negative being 0. This is due to the different mapping in the training process.

IV. Results and Discussion

A descriptive and comparative analysis between the two IndoBERTweet-based predictions and the manually assigned sentiment labels, along with the results and their implications, will be discussed in detail in this chapter.

A. Results of Manual Labeling

After manually labeling the data into three sentiment classes—positive, negative, and neutral—the results were grouped based on the originating platforms of Twitter, Instagram, and Tiktok. This grouping aimed to examine how sentiment distribution varies across different social media environments. For each platform, the proportion of sentiments was calculated and visualized using table and bar chart.

As shown in Table III and Fig. 3, the sentiment distribution of Twitter, Instagram, and Tiktok comments shows a relatively similar pattern with negative sentiments dominating.

TABLE III
Table of Manual Labeled Sentiments Distribution by Platform

Label	Twitter	Instagram	TikTok
Negative	51.72%	51.62%	69.39%
Neutral	33.23%	32.19%	21.84%
Positive	15.05%	16.19%	8.78%

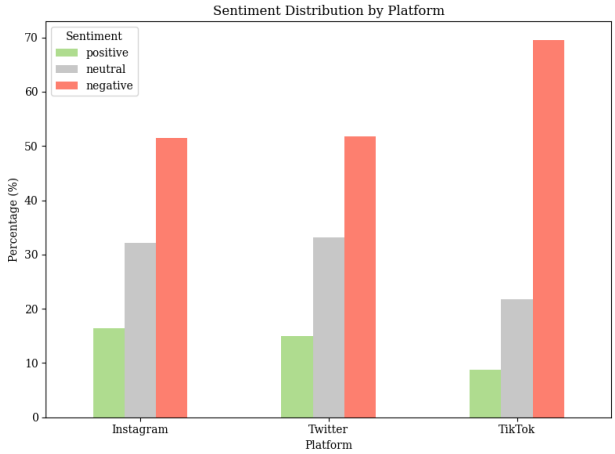

Fig. 3. Bar Chart of Manual Labeled Sentiments Distribution by Platform

The dominance of negative comments is likely due to the hashtag "#KaburAjaDulu", which is commonly used to criticize the political system. Many posts and videos featuring this hashtag express dissatisfaction or frustration, thus shaping the negative responses. Since criticism is categorized as negative, a higher percentage of negative sentiment was found.

TikTok exhibits the highest percentage of negative comments, while Twitter shows the least, likely due to demographic differences. This might be due to the majority of TikTok users being young women and Twitter being older men, as found in Chapter II. Younger users tend to imitate sarcastic tones without full context, whereas older users are more thoughtful or neutral. Additionally, women have greater tendency for self-expression compared to men, leading to more active commenting.

Platform design and content formats further influence interaction styles. Twitter, which is based on short text posts with a limit of 280 characters [41], encourages concise, sharp, and often sarcastic remarks. Moreover, since replies are public, users may argue more just to be seen by others. Instagram, on the other hand, promotes more positive and aesthetic-centered interactions due to its focus on visual elements. Meanwhile, TikTok's short video format, autoplay, and fast-scrolling feeds stimulate quick emotional responses.

B. Results of Hyperparameter Tuning and Transfer Learning

Table V summarizes the results of hyperparameter tuning. Lower training loss and higher F1-score indicate better models. As can be seen, higher epochs (5–6), higher learning rate, and lower weight decay (0.02-0.15) boosts performance.

The best model is not the one that fits training data best, but the one that performs best on unseen data. That was why the evaluation metrics chosen was F1-score, not training loss. F1-score evaluates precision and recall on the validation data.

The best-performing trial is Trial 2 which achieved an F1-score of 0.740. It is then implemented as follows: learning rate = 3.5e-5, epoch = 6, weight decay = 0.025, scheduler = linear, and batch size = 4. When evaluated with the test dataset, the model achieved an F1-score of 0.745 and an accuracy of 0.75. This model will now be called Custom IndoBERTweet.

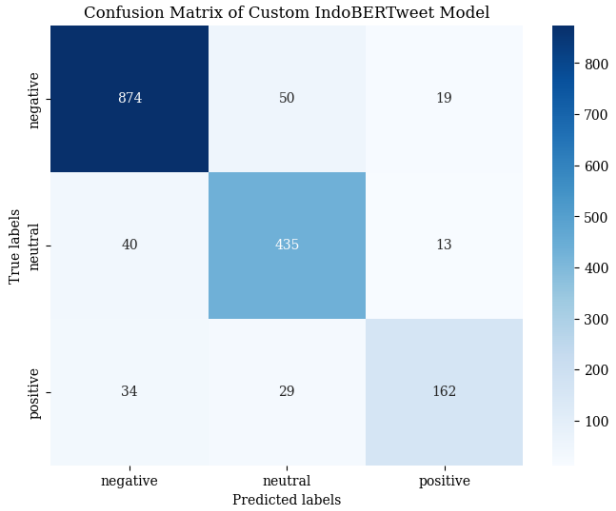

Fig. 4. Confusion Matrix of Custom IndoBERTweet

TABLE IV
TABLE OF THE CONFUSION MATRIX RESULT

Sentiment	True Positive	Misclassifications
Negative	874	As Neutral: 50 & Positive: 19
Neutral	435	As Negative: 40 & Positive 13
Positive	162	As Negative: 34 & Neutral 29

Table IV and Fig. 4 provide insight into the Custom IndoBERTweet's performance per sentiment class using confusion matrix, while Fig. 5 shows the ROC curve which plots True Positive Rate (TPR) against False Positive Rate (FPR) for each class. An AUC (Area Under Curve) close to 1 indicates excellent predictive performance.

Both the confusion matrix and ROC curve show that the Custom IndoBERTweet model performs better on negative and neutral sentiments than on the positive. This is probably due to the high amount of negative comments and few amount of positive ones in the dataset, resulting in the model not knowledgeable enough to classify positive sentiments. Positive feelings are also often expressed by non-text ways, like emojis or GIFs. The model can also be generalized, meaning that it lazily predicted most comments as negative, especially because negative sentiment dominates the training data.

C. Comparison Between Automatic and Manual Labeling

The two approaches of automatic sentiment classification were using the fine-tuned IndoBERTweet and Custom IndoBERTweet models. The results will be compared to each other, as well as to the manually labeled sentiments.

Table VI presents the sentiment distribution across platforms with different sentiment classification methods. The Custom IndoBERTweet model offers more balanced and accurate sentiment classification across platforms compared to the original fine-tuned model.

TABLE V
PERFORMANCE OF INDOBERTWEET ACROSS HYPERPARAMETER TUNING TRIALS DURING TRANSFER LEARNING

Trial	Steps	Learning Rate	Epochs	Weight Decay	Scheduler	Batch Size	Training Loss (on 500th step)	F1-score
1	747	2.06e-5	3	0.213	linear	4	0.697	0.739
2	**1494**	**3.37e-5**	**6**	**0.025**	**linear**	**4**	**0.196**	**0.740**
3	1494	4.99e-5	6	0.149	linear	4	0.083	0.737
4	500	2e-5	4	0.183	cosine	8	0.015	0.723
5	625	1.17e-5	5	0.000	linear	8	0.000	0.729
6	1245	1.77e-5	5	0.081	linear	4	0.003	0.688
7	1245	1.35e-5	5	0.271	linear	4	0.000	0.713
8	750	1.03e-5	6	0.191	linear	8	0.007	0.711
9	996	2.5e-5	4	0.274	linear	4	0.000	0.720
10	500	1.61e-5	4	0.023	cosine	8	0.004	0.711

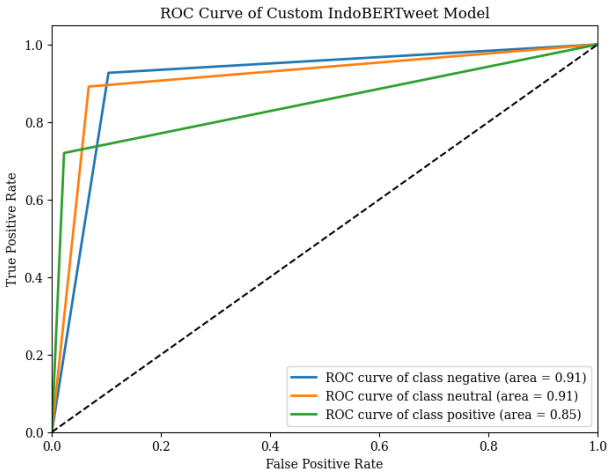

Fig. 5. ROC Curve of Custom IndoBERTweet

TABLE VI
COMPARISON OF SENTIMENT DISTRIBUTION BY PLATFORM: INDOBERTWEET VS MANUAL LABELING

Platform	Model/Method	Negative	Neutral	Positive
Twitter	Fine-tuned	55.96%	25.55%	18.50%
	Custom	51.57%	36.68%	11.76%
	Manual Labeling	51.72%	33.23%	15.05%
Instagram	Fine-tuned	57.90%	16.38%	25.71%
	Custom	53.52%	32.19%	14.29%
	Manual Labeling	51.62%	32.19%	16.19%
TikTok	Fine-tuned	55.71%	17.55%	26.73%
	Custom	68.57%	22.65%	8.78%
	Manual Labeling	69.39%	21.84%	8.78%

Table VII also shows that the Custom IndoBERTweet model significantly outperforms the original fine-tuned version in terms of accuracy and weight F1-score. This is possibly because the Custom IndoBERTweet model was fine-tuned with domain-specific data, allowing it to better recognize and understand slang, colloquial expressions, and platform-specific context. The highest accuracy, 90.63%, was achieved on TikTok, likely because the platform had the most negative comments. As found before, the Custom IndoBERTweet model classifies negative sentiment with greater accuracy.

TABLE VII
COMPARISON OF MODELS ACCURACY BY PLATFORM

Platform	Model/Method	Accuracy	F1-Score (weighted)
Twitter	Fine-tuned	66.51%	0.66
	Custom	86.85%	0.87
Instagram	Fine-tuned	61.03%	0.60
	Custom	89.54%	0.89
TikTok	Fine-tuned	60.29%	0.64
	Custom	90.63%	0.91

V. CONCLUSIONS AND FUTURE WORKS

This study analyzes and compares public sentiments in Indonesian-language comments on Twitter, Instagram, and TikTok using IndoBERTweet, focusing on the viral hashtag "#KaburAjaDulu" ("#JustRunAwayFirst"). A total of 1,659 preprocessed user comments were classified based on their sentiments. The results show a dominance of negative sentiments, highlighting how socio-political issues influence emotional expression on social media.

To improve accuracy in sentiment classification, the pretrained IndoBERTweet model was fine-tuned using transfer learning with a manually labeled dataset. The result, Custom IndoBERTweet model, achieves an F1-score of 0.745 and accuracy of 75%, outperforming the original pretrained fine-tuned model on the same dataset. This demonstrates the effectiveness of transfer learning in adapting pretrained models to specific contexts for real-world applications.

This approach has limitations, including reliance on active users, limited data, the lack of labeling expertise, and sampling bias from a single hashtag. Future research should explore multimodal data, passive interactions, or broader opinions and topics. Manual labeling should also be conducted by specialists to improve accuracy and reduce bias.

VI. CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Stefani Gifta Ganda: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Visualization. **Edeline Kenni**: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Visualization. **Nicholaus Hendrik Jeremy**: Supervision. **Rhio Sutoyo**: Conceptualization, Validation, Resources, Writing - Review Editing, Supervision.

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